Mudit Atrey

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Education

Kanpur Institute of Technology, AKTU — Kanpur, India
Bachelor of Technology in Computer Science and Engineering (AI/ML Specialization) — *2023 - June 2027*

- **Academic Standing:** CGPA 7.3/10.0; Core Focus: Machine Learning, Numerical Methods, Scientific Computing.
- **Availability:** Available immediately for full-time on-site or remote engineering roles (College NOC provided; end-semester exams only).

Autonomous Systems, GNC & Simulation

Physics-Regularized Geometric Manifolds for Surrogate-Assisted MDO of Tailless BWB UAVs — Aug 2026

- **Publication Status:** Sole-author preprint (DOI: 10.5281/zenodo.22893593); under external peer review at *Aerospace Science and Technology* (**Elsevier, IF 6.4, Q1**), Manuscript ID: AESCTE-D-26-06017.
- Formulated a C^1-continuous Hermite geometric manifold and 5-MLP surrogate ensemble with mass-centroid balancing; active learning increased feasibility from 28.1% to 54.7% and delivered a +7.5% (+20.9 km) range gain across 10,087 evaluations.
- **3D RANS CFD Cross-Check (Additional to Submitted Manuscript):** Executed SimScale/SU2 RANS (Spalart-Allmaras) on representative BWB loft; isolated winglet junction corner separation and 3D turbulent boundary-layer accumulation absent in strip methods.
- **OpenCASCADE B-Rep Pipeline:** Built custom B-Rep lofting pipeline on OpenCASCADE to eliminate micro-gaps and zero-thickness trailing edges from polygon translations, producing certified watertight solids for CFD meshing.

6-DOF Aerodynamic Guidance & Trajectory Simulation (SIH26) | *C++, Python, EKF, Onshape* — 2026

- Modeled 6-DOF ballistic trajectory simulation and Augmented Proportional Navigation (APN) guidance law for 155mm munition, targeting $\leq$ 30m CEP over 20km+ ballistic arc (Ministry of Defence / SIH26098).
- Implemented 16-state Extended Kalman Filter (EKF) fusing 500Hz IMU with 10Hz GNSS; designed cruciform canard array decoupling passive roll despin from pure lateral steering forces.
- Demonstrated in closed-loop 7-DOF simulation (7th DOF modeling roll-isolated collar dynamics); cut spin from 300 rps to $<$ 5 rps with 2.34m CEP under IMU/GNSS sensor noise injection.

Experience

Pradite (pradite.com) — Kanpur, India
Founder & Technical Lead — *Aug 2026 - Present*

- Founded zero-knowledge BYOK AI coding platform; led 5-person engineering team through full-stack architecture, security core, and Early Access launch.
- Engineered zero-trust cryptographic core: client-side XChaCha20-Poly1305 + Argon2id encryption (libsodium), sub-5ms encrypted IndexedDB index search, and direct client-to-model streaming with zero server-side storage.
- Built multi-provider inference engine with dynamic complexity routing across 17 APIs/runtimes (vLLM, Ollama); filed Class 42 trademark (IP India) and executed pre-incorporation co-founders agreement.

Software Engineering & Open Source

scikit-learn Ecosystem (probabl-ai / skrub-data) — Open Source
Open Source Contributor (ESoC) — *2025 - Present*

- **skrub:** Refactored CheckInput pipeline logic to bypass eager evaluation of Polars lazyframes, cutting memory overhead during dataset ingestion.
- **skore:** Implemented multi-metric aggregate parameters for ImpurityDecreaseDisplay and CoefficientsDisplay, enabling flexible model explainability for tree and linear ensembles.
- Fixed figure-level legend persistence during file export in PredictionErrorDisplay; 5 merged PRs reviewed and approved by core maintainers (*@glemaitre*, *@rcap107*, *@GaetandeCast*).

Projects

LLM-Assisted Parametric CAD Framework (onshape-mcp) | *GitHub* | *Python, MCP* — 2026

- Extended an open-source MCP foundation into a 69-tool agentic CAD system driving programmatic 3D geometry, feature trees, and parametric editing in Onshape via REST API.
- Engineered multi-modal verification loop and empirical benchmark (1.000 accuracy on human specs vs 0.533 on drawing extraction); covered by 698 unit tests.

Technical Skills & Interests

Languages: C++, Python, Rust, TypeScript, JavaScript, SQL, Linux Shell / Bash

Robotics, GNC & CFD: 6-DOF Trajectory Sim, 16-State EKF Sensor Fusion, APN Guidance, SimScale (RANS CFD), SU2, OpenCASCADE, Onshape API, AVL, XFOIL

Data Science & ML: PyTorch, scikit-learn, Optuna, NumPy, SciPy, Pandas, MLX, Physics-Informed ML

Systems & Hardware: Git, GitLab CI/CD, Docker, Linux, WebAssembly, Supabase, PlatformIO, ESP8266

Interests: Autonomous UAVs, Defense Technology, Chess (1600 ELO), Scientific Computing